

Different program design strategies

- ~~3D method~~ Greedy Method } used when we need to optimise current solⁿ maximize profit, minimize loss
- Dynamic programming
- Divide & Conquer

Greedy Method

Greedy method ek algo create krte h & assume krte h ki jo ye method h wo best optimized h.

& use pr work krte h, wo aag aag possibility se try krte krte

Brute force approach $\Rightarrow$ Try all the possibilities & select best option

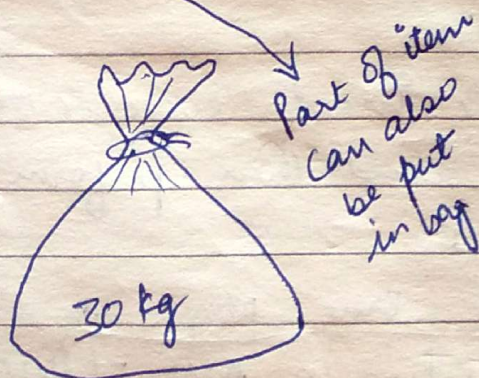
Eg ~~arrange~~ sort n elements

Try all combinatⁿ = $n!$ options
unme se 1 sorted krni hoga
but $O(n!)$ is intractable = ∞ time

Dynamic ~~method~~ program hui brute ke tarh h
hudi hudi subproblems bnati h
& unme se best possibility pr work krte h

Fractional Knapsack problem

we are having a bag, we are putting items in bag. On putting items, we get certain profit. finding a way to maximize profit, is required.



Items $\rightarrow$	I1	I2	I3	I4
Weight	10	20	10	20
Profit	10	20	20	40
P/W	1	1	2	2

Complexity of Fractional knapsack problem

for ($i = 0$ to n)
{ $a[i] = \frac{P[i]}{W[i]}$ } $\rightarrow O(n)$

}

sort $a[i]$ $\rightarrow O(\log n)$

$i = 0$

while knapsack is not full $\rightarrow O(n)$

{

put ^{item} ~~item~~ with highest P/W

$i++$

}

$\Rightarrow$ complexity = $O(n \log n)$

Using Greedy approach

Find item with highest Profit ratio
weight

Fill bag with those items with highest ratio

⇒ Put $I_4 + I_3$ in bag
Total weight = 30 kg
Total Profit = 60

Greedy Strategy

Here we used a strategy & solved the ques
whereas Dynamic Prog use krte to har item ko put
kr - kr k dekhte.

Zero-one knapsack problem

Ya to poora item sack me jaega, ya jaega hi nai.



	I_1	I_2	I_3	I_4
w	25	10	10	10
P	50	19	19	19
P/w	2	1.9	1.9	1.9

Yaha agar Greedy approach lagai to sbse phle
25 kg wala bag me rkha
Pr 5 kg bag khali reh gya
Total profit = 50

Instead agar 10, 10, 10 kg k rkhte bag me
then total profit = $19 + 19 + 19 = 57$

So Greedy cant be used for 0-1 knapsack problem

2nd Greedy Problem Huffman's Code / Prefix code

Jin characters ki freq jyada unke code ki length kam kr do

Eg In english AEIOU ki freq sbse jyada hai to unki code length kam kr do

A - 0

E - 1

I = 00

?

00

aga wo I hai ya AA hai?

This problem occurs coz one code is prefix of other code

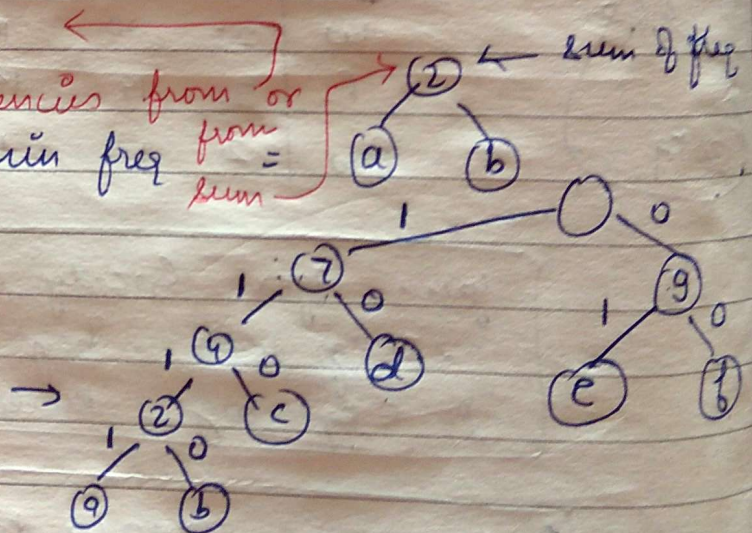
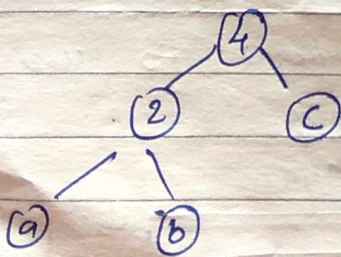
Instead use Huffman codes or prefix codes

No code is prefix to other code

a b c d e f

freq → 1 1 2 3 4 5

⊕ Connect least 2 frequencies from or find elements of min freq from sum



a = 1111

d = 10

b = 1110

e = 01

c = 110

f = 00

Say Sender sends 00110011110
b e e b

fceb

Greedy Problem

Optimal Merge Pattern

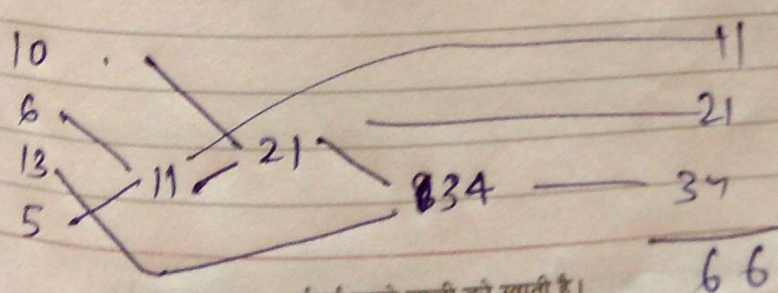
we have files of different sizes
 size is given in records
 Merge the files such that lowest no
 of records are to be handled

file	size	merge
1	10	} 16
2	6	
3	13	} 29
4	5	
		} 34
		79

10	} 34	34
6		29
13	} 18	18
5		74

Solution $\Rightarrow$ Merge smaller files first

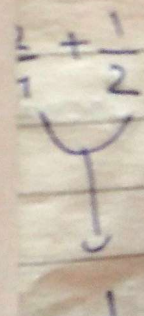
better approach



2015

ईर्ष्या करने वाली को खाती है।
 Envy devours him who harbours it.

Min



20
14
5-11

MARCH
WEDNESDAY
Week 11 (070-295)

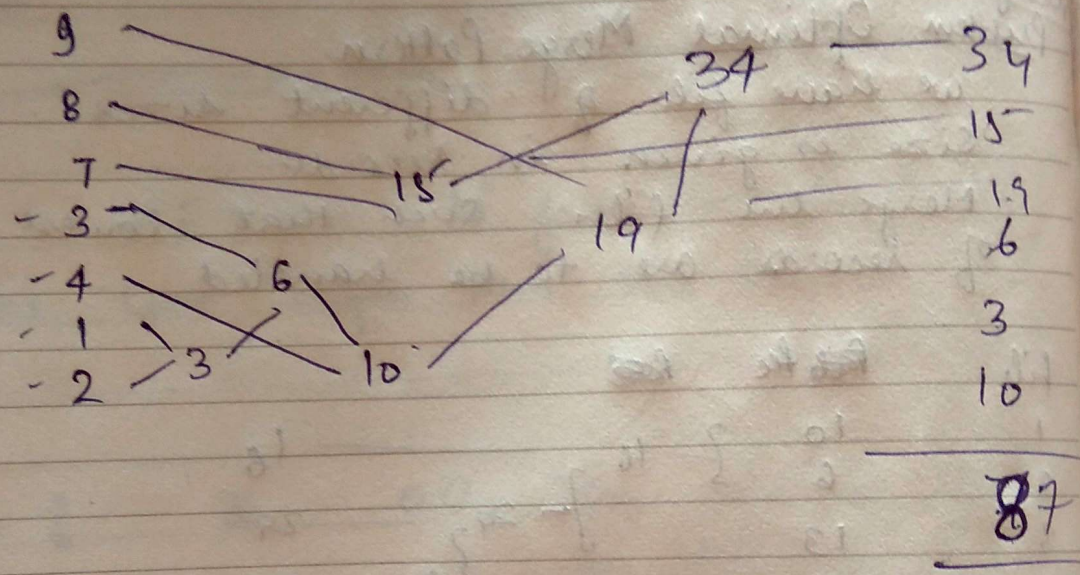
मार्च
बुधवार
सप्ताह ११ (०७०-२९५)

विक्रम सम्वत् २०७१
चैत्र कृष्ण १०
शक सम्वत् १९३६
फाल्गुन २०

Vikram Samvat 2071
Chaitra Krishna 10
Shak Samvat 1936
Phalgun 20

सूर्योदय ६:०६ Sun Rise 6:09

सूर्यास्त ५:५९ Sun Set 5:59

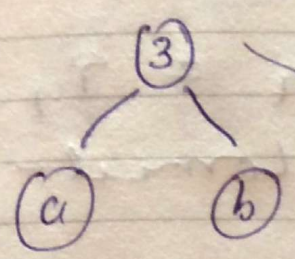


2015

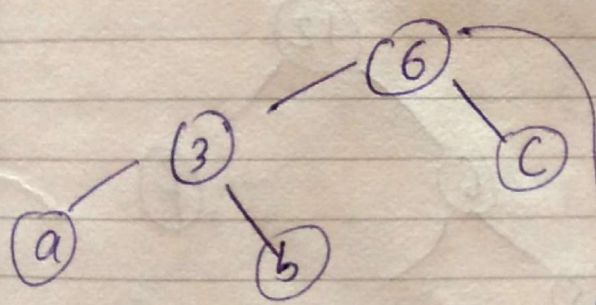
आश्रय और बल है एवं आपत्ति के समय में रक्षा करता है।
age and strength, a very present help in trouble.

a b c d e f
 1 2 3 4 5 6

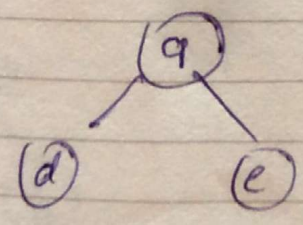
Min freq = a, b $\Rightarrow$ connect them
 1 2



Now min freq = 3 & c(3) $\Rightarrow$ connect them



Now min freq = d & e $\Rightarrow$ connect them



Now min freq = 6 & f(6)
 connect them

2015

"मात्रा स्पर्श आते हैं, जाते हैं। उसे सहन करो।"
 "Contacts of the senses with their objects come and go, endure them."

01

Y9/MARCH

Y/SUNDAY

Week 09 (000-305)

मार्च

रविवार

सप्ताह ०९ (०६०-३०५)

विक्रम संवत् २०७१

फाल्गुन शुक्ल ११

शक संवत् १९३६

फाल्गुन १०

Vikram Samvat 2071

Phalgun Shukla 11

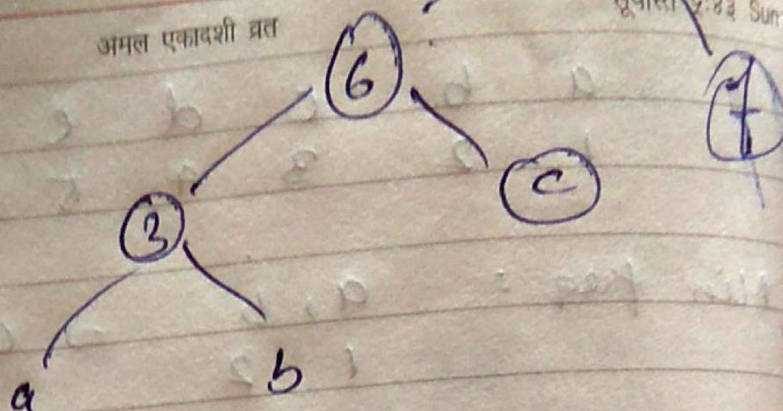
Shak Samvat 1936

Phalgun 10

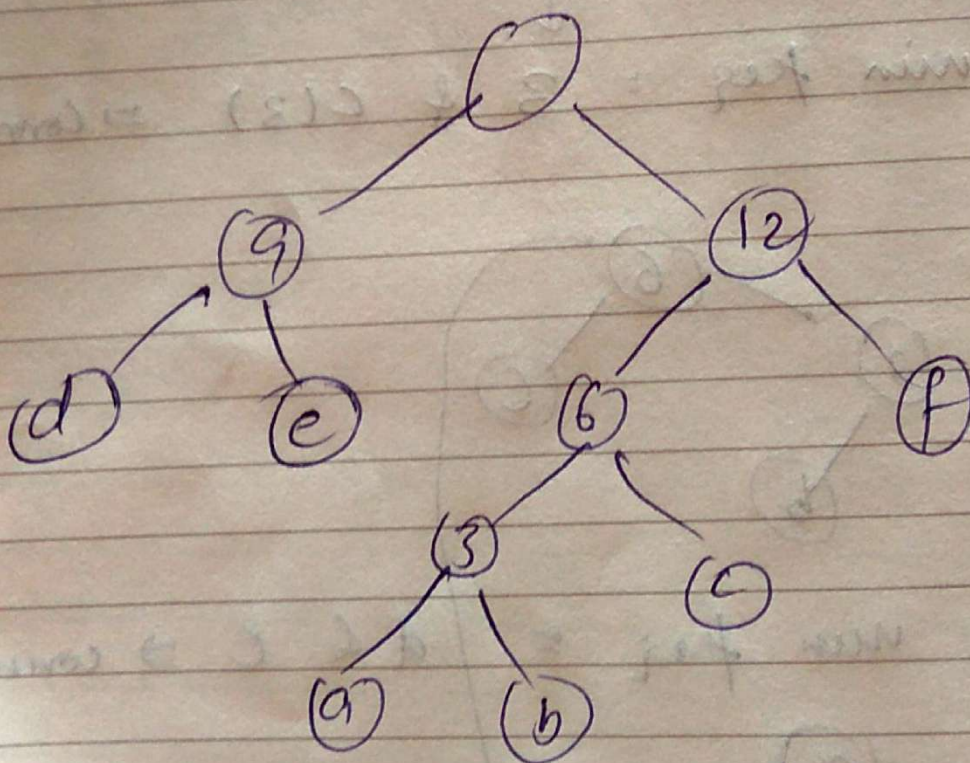
सूर्यास्त ५:४३ Sun Set 5:43

सूर्योदय ६:१७ Sun Rise 6:17

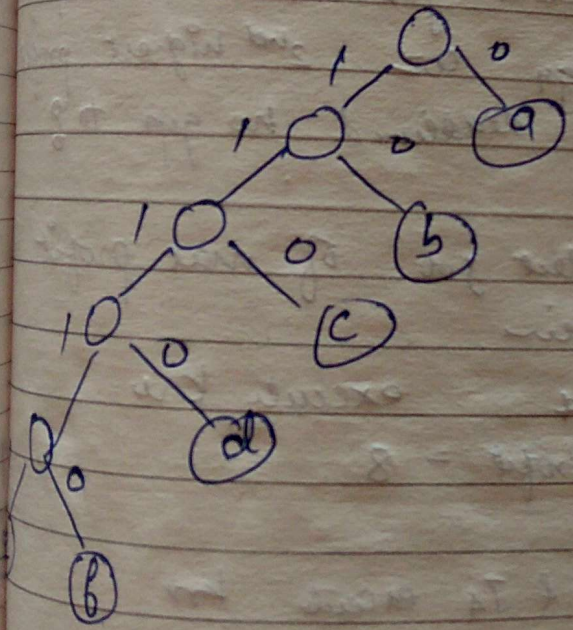
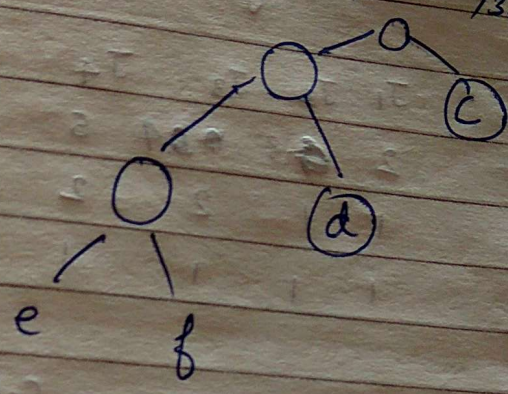
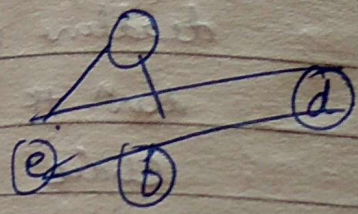
अमल एकादशी व्रत



Now least 2 free = 9, 12
Lowest



a b c d e f
 $\frac{1}{2}$ $\frac{1}{4}$ $\frac{1}{8}$ $\frac{1}{16}$ $\frac{1}{32}$ $\frac{1}{32}$



a = 0
 b = 10
 c = 110
 d = 1110
 e = 11110
 f = 11110

Avg length = $\frac{5}{32} + \frac{5}{32} + \frac{4}{16} + \frac{3}{8} + \frac{2}{4} + \frac{1}{2}$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 0.15625 0.25 0.375 1

$\sim$
 0.3125

71

Minimum

Put jobs with highest Profit just before its deadline
 J₁ 6 5
 J₂ 5 4

Put it just before 5
 2 to n.

3rd Greedy Prob Job scheduling with deadline

4 Jobs	J ₁	J ₂	J ₃	J ₄
Profit	2	3	4	5
deadline	1	1	2	2
burst	1	1	1	1

deadline means is time
 durⁿ + wo khatam hoo
 chae

1st Greedy strategy = Select job with max profit
 innappropriate strategy coz 2nd highest profit
 wala job agar deadline ho gya to ?
 FAIL!

2nd Greedy strategy : Select job of max profit with
 nearest deadline
 J₂ & J₄ execute kro
 ⇒ Profit = 8
 FAIL!

but
 actual ans = J₃ & J₄ execute kro
 Profit = 9

So Greedy algo fails for Job Scheduling with
 deadline problem

It can't even be solved with Dynamic prog

⇒ NP Complete problem ⇒ algo's to hai pr
 takes years to solve

NP Hard Problem ⇒ Can't be solved

Vikram Samvat 2071
Chaitra Krishna 3
Shuk Samvat 1936
Phalgun 17

Vikram Samvat 2071
Chaitra Krishna 3
Shuk Samvat 1936
Phalgun 17

मार्च
रविवार

MARCH
SUNDAY

08

सप्ताह 90 (050-055)

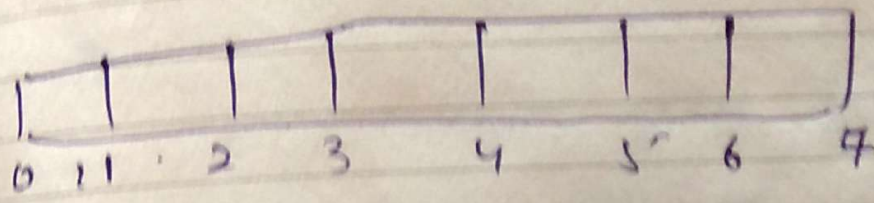
Week 10 (067-298)

सूर्योदय 5:44 Sun Set 5:49

सूर्योदय 5:49 Sun Rise 6:11

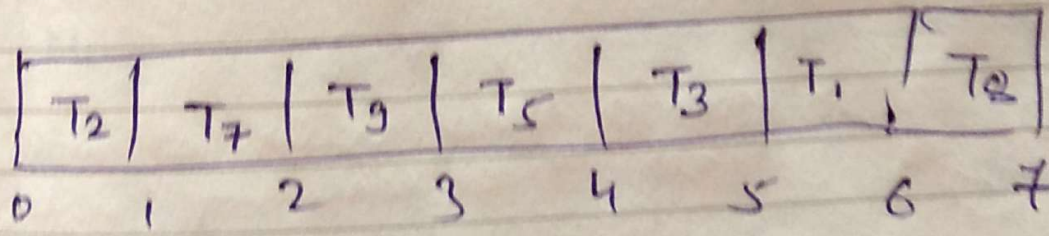
CSOS
84

Task	T ₁	T ₂	T ₃	T ₄	T ₅	T ₆	T ₇	T ₈	T ₉
Prof	15	20	30	18	18	10	25	16	25
Dead	7	2	5	3	4	5	2	7	3



Deadline is 7 and 9 jobs are there
so 2 jobs will be left out

T₃ has Max Profit Put it just before
its deadline = 5



T₄ & T₆ are left out

Profit = 147

100
16
25
147

2015

किस्मत पर सब न छोड़ें, न पुरुषार्थ का फांका करें।
Let us not leave everything to destiny, nor be vain about our endeavour.

Prim's & Kruskal algo (for min spanning tree) can be solved using Greedy algo.

One max prob $\Rightarrow$ Sequential Magnetic tape ^{prob}
To assign files on a sequential magnetic tape

Save shortest files first so as to ~~require~~ less time is required to access files.

Files of different sizes.

File	F ₁	F ₂	F ₃	F ₄
Size	10	20	5	2

Optimal Merge Pattern $\longleftrightarrow$ Greedy Problem

Merge the files so that least no of records are to be handled

$$\begin{aligned}
 &\text{Merged } F_1 F_2 \rightarrow 30 \\
 &\quad + F_3 \\
 &\quad \quad \underline{35 + F_4} \\
 &\quad \quad \quad 37
 \end{aligned}$$

$$\begin{aligned}
 \text{Total records} &= 30 + 35 + 37 \\
 &= 102 \text{ records}
 \end{aligned}$$

$$\begin{array}{rcl}
 F_4 & 2 & \} \quad 7 \\
 F_3 & 5 & \} \quad 27 \\
 F_2 & 20 & \} \quad 37 \\
 F_1 & 10 &
 \end{array}$$

$$\text{Total} = 7 + 27 + 37 = 71$$

Optimal merge pattern says Merge smaller files first

$$\begin{array}{rcl}
 F_4 & 2 & \} \quad 7 \\
 F_3 & 5 & \} \quad 17 \\
 F_1 & 10 & \} \quad 37 \\
 F_2 & 20 &
 \end{array}$$

$$= 7 + 17 + 37 = \boxed{61} \text{ Minimum}$$

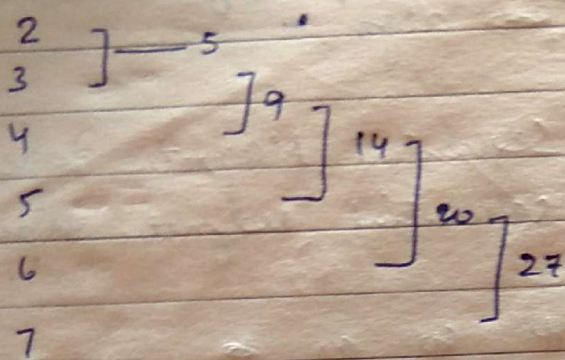
Edge relaxation =

To relax each element connected directly to it

$$\text{dist } B = \min \left(\underset{\substack{\uparrow \\ \text{old dist from source vertex}}}{\text{dist}(B)}, \underset{\substack{\uparrow \\ \text{weight}}}{\text{dist}(A) + w(A, B)} \right)$$

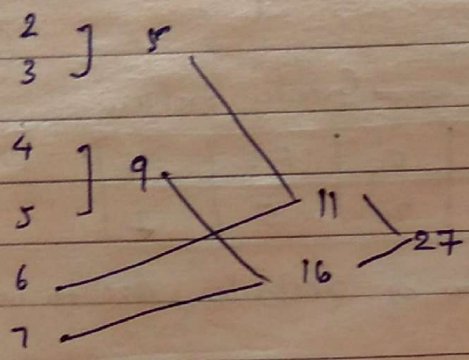
Roll No.
Date

Q Minimum record pte req to merge these files 2 3 4 5 6 7



X wrong

= 75 records (min)

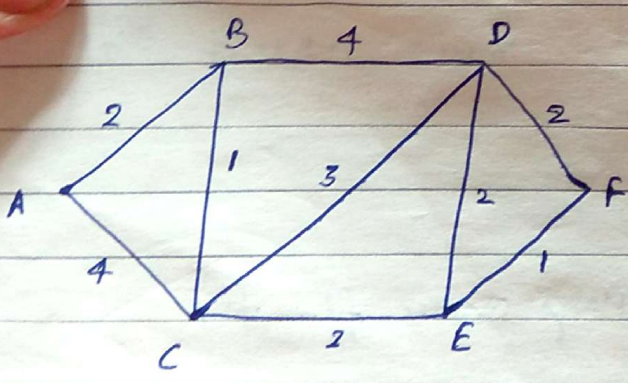


$$= 5 + 9 + 11 + 16 + 27 = 228$$

given a source vertex, it finds shortest path to all remaining vertex.

Dijkstra Problem (Single source shortest path algo)
Given the source vertex, It find shortest path from remaining vertices

* Works well on both directed & undirected graph



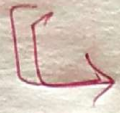
Creates 2 Sets

Set of visited vertex
& set of unvisited vertex

Contains vertices whose shortest path is determined by algo

whose path is yet to be discovered

In each iteration we have to visit a new vertex



Out of 0, 2, 4, ∞ - min = 2 $\Rightarrow$ Mark it visited
Now perform edge relaxation on B

Iteration 1

	visited	distance	Through
A	✓	0	-
B		2	A
C		4	A
D		∞	
E		∞	
F		∞	

A

unvisited me
shortest
job hi
in next
visited
mean to

	visited	Dist	Thru
A	✓	0	-
B	✓	2	A
C		(2+1)=3	B
D		(2+4)=6	B
E		∞	-
F		∞	-

AB

A	✓	0	-
B	✓	2	A
C	✓	3	B
D		8+3=6	C
E		3+2=5	C
F		∞	

ABC



A	✓	0	-
B	✓	2	A
C	✓	3	B
D		5+2	E
E	✓	5	C
F		(5+1)	E

ABCE



A	✓	0	-
B	✓	2	A
C	✓	3	B
D	✓	6	B
E	✓	5	C
F	✓	6	E

ABCDEF

Shortest path

Find shortest path from A - F

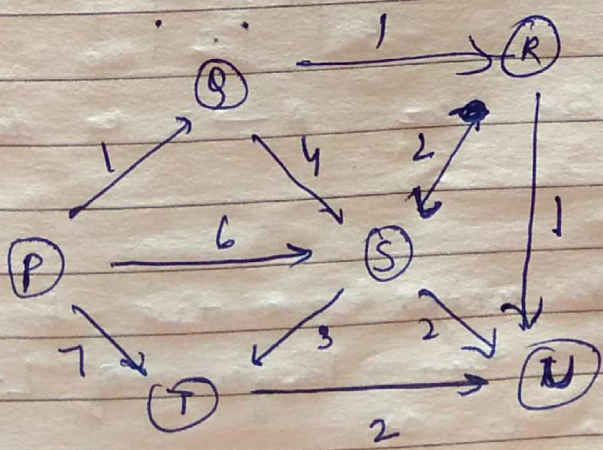
A $\leftarrow$ B $\leftarrow$ C $\leftarrow$ E $\leftarrow$ F

$\Rightarrow$ A-B-C-E-F is

Shortest dist b/w A-F = 6 = Entry in dist column

Shortest path

CSOY
44



If there are n vertices then n-1 iterations are req

P	✓	0	-
Q		①	
R		∞	
S		6	
T		7	
U		∞	

P

	✓	0	-
	✓	1	P
		②	
		5	
		7	
		∞	

PQ

	✓	0	-
	✓	1	P
	✓	2	Q
		4	
		∞	
		③	

PQR

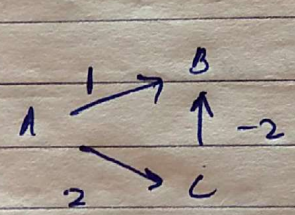
P	✓	0	
Q	✓	1	P
R	✓	2	Q
S		④	
T		∞	
U	✓	3	R

PQRU

PQRUS

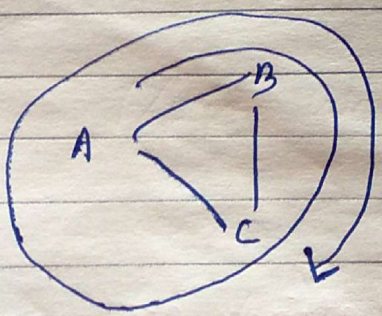
PQRUST

BUT EDGE WEIGHTS MAY BE NEGATIVE TOO



A	✓	0	A
B		①	
C		2	

So dijkstra fails here coz $A \rightarrow B$ ka dist 0 hai



If edge weight are +ve & graph is cyclic then length of Longest path is ∞

$A \rightarrow C \Rightarrow A-B-C-A-B-C \dots = \infty$

Vikram Samvat 2071
Chaitra Krishna 6
Shuk Samvat 1936
Phalgun 21

विक्रम संवत् २०७१
चैत्र शुक्ल ६
शुक्र संवत् १९३६
फाल्गुन २१

Belman ford मार्च

गुरुवार

सप्ताह ११ (०७१-२६४)

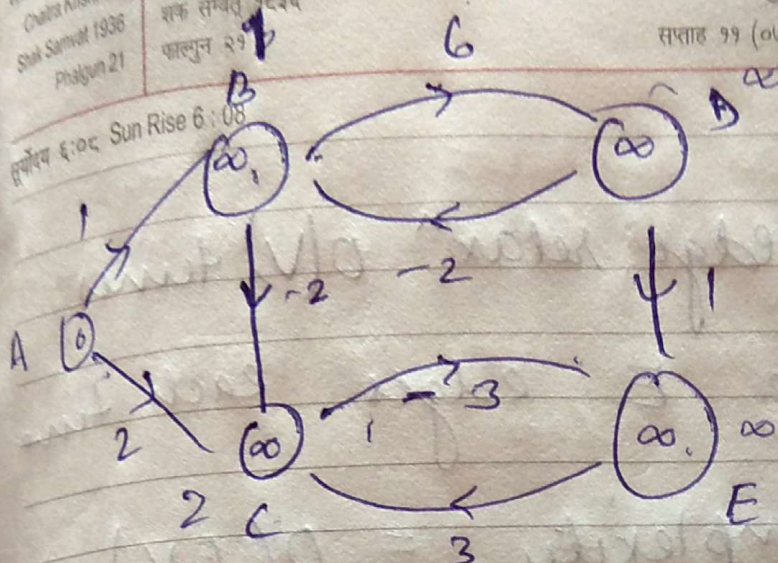
MARCH

THURSDAY

Week 11 (071-294)

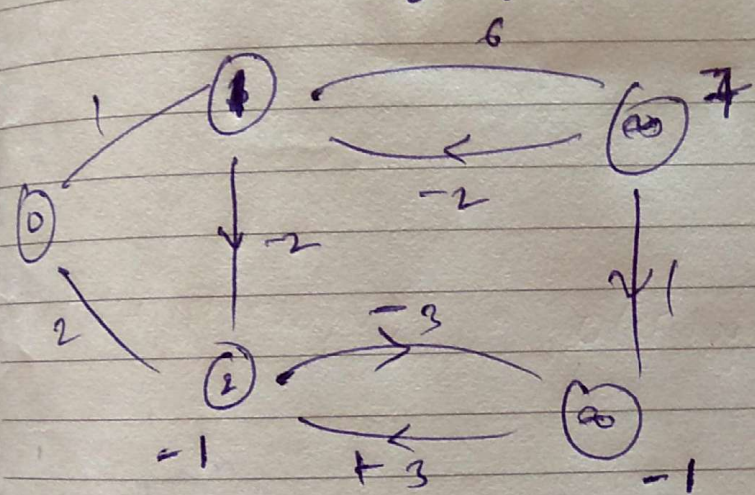
12

सूर्यास्त ५:५२ Sun Set 5:52



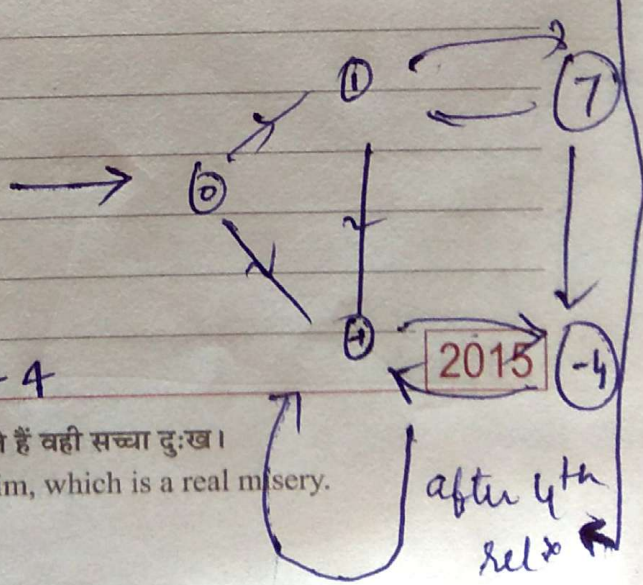
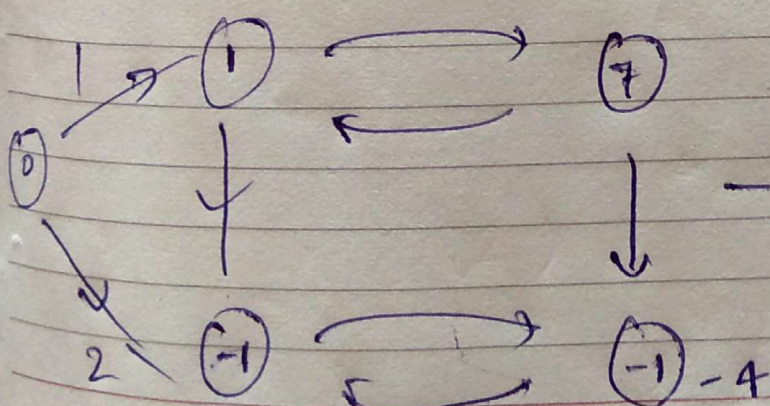
5th relaxⁿ
pr koi change
nhi
ana
chahiye!

↓ after 1st relaxation



4th pr hi
koi change
nhi
aaye

↓ after 2nd relⁿ



after 4th
relⁿ

ईश्वर हमको कभी नहीं भूलता, हम भूलते हैं वही सच्चा दुःख।

God never forgets us it is we who forget him, which is a real misery.

We perform edge relaxⁿ $O(V)$ times
& see E edges each time

$$\Rightarrow \text{Complexity} = O(EV)$$



Kram Samvat 2071
hakra Krishna
hak Samvat 1936
hailgun 22

विक्रम सम्वत् २०७१
फाल्गुन शुक्ल ८
शके सम्वत् १९३६
फाल्गुन ७

फरवरी

गुरुवार

सप्ताह ०९ (०५१-३०८)

FEBRUARY

THURSDAY

Week 09 (057-308)

26

सूर्योस्त ५:४१ Sun Set 5:41

Dijkstra

Initialize the graph
Create Min heap of all the objects & its name is Q // order(v)
while (Q != null) {
 Extract
 x = ~~delete~~ Min(Q); // O(log v)
 For edges thru x which are unvisited perform edge relaxation;
 Reheapify(Q); O(E log v)
}

$$(v \log v) + O(E \log v) = O(E \log v)$$

After each edge relaxation we need to ~~store~~ reheapify it coz weight change ho jaenge

if we use array in place of heap, then complexity = v^2

Coz extract min operation will be O(v) & while loop runs for v times

$$\Rightarrow O(v^2)$$

We can't use sorted array coz after each delete min operation, we need to sort the array again

In array implementation, edge relaxation

we take constant time, we need need to store the new value,
मिथ्या ज्ञान से हम हमेशा डरते रहें।
Let us always beware of false knowledge.

21

FRIDAY

Week 09 (058-307)

शुक्रवार

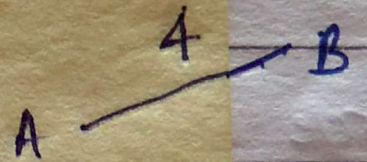
सप्ताह ०९ (०५८-३०७)

सूर्योदय ६:१८ Sun Rise 6:18

सूर्यास्त ५:४२ Sun Set

If our graph has no possibility of -ve edges. Eg ~~the~~ computer networking at such places we go with dijkstra algo as compared to Bellman Ford

Eg
If there exist shortest path in the graph then after $(n-1)$ relaxations we will get the shortest paths & at n^{th} relaxation, there will be no change in the graph & if n^{th} relaxation we blie weight change ho gya, then there exist no shortest path $\Rightarrow$ There exist ~~no~~ cycle of -ve edges.



e edge
do

A
B
C
D

If all edges are -ve & graph is cyclic then dijkstra definitely fail
 but if there exists no cycle of -ve edges, then dijkstra may work or may fail.

If there exist cycles of -ve edges in graph then min path path can't be found

* Dijkstra is Greedy algo with some dynamic programming

* ~~Best~~ Advanced Dijkstra $\Rightarrow$ Bellman Ford

$\hookrightarrow$ Purely dynamic programming

* Dijkstra perform edge relaxation for unvisited vertices only

* Bellman Ford all

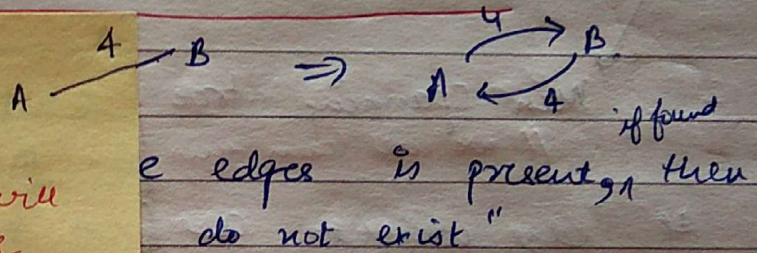
Greedy prob may not produce optimum solⁿ but dynamic prog sol will definitely be optimum

Bellman Ford Algo (* Must need a directed Graph)

$\hookrightarrow$ performs edge relaxation $n-1$ times

① If Graph is undirected convert to directed

If there exist shortest path in the graph then after $(n-1)$ relaxations we will get the shortest paths & at n th relaxation, there will be no change in the graph & if n th relaxation we find weight change has gone, then there exist no shortest path $\Rightarrow$ There exist no cycle of -ve edges.



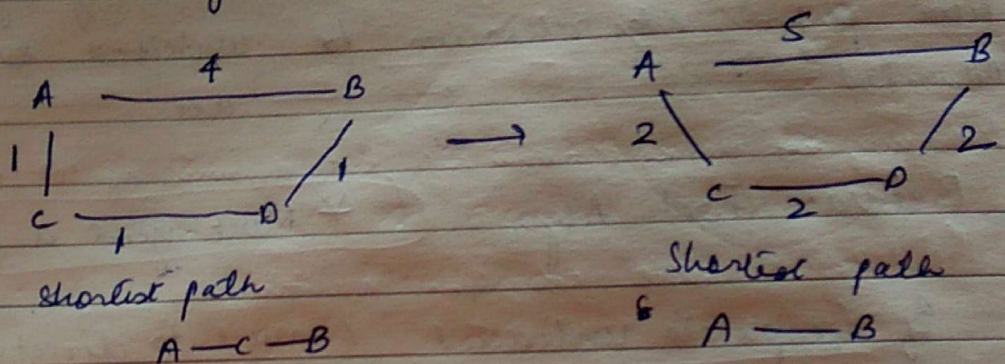
	Dist	Thru	
A	0		
B	1	A	
C	4	BA	let the B now
D	∞		
E	∞		

* Check dist of B to all remaining

* Then check dist of C with all remaining
 say dist of B reduced due to above step
 then consider that vertex again by going to step

Complexity of Dijkstra = $O(V \log E)$
 or Bellman Ford = $O(VE)$

Q If all edge weights are changed by constant factor (say 5) then will the shortest path change or not?
 Sol It will change



Min spanning tree same if all edge weight badhane pr
 // we will learn this in discrete.

Q If edge weight are scaled by some factor (multiplied) then shortest path will change or remain same?
 Ans = remain same

Best way to find shortest path in unweighted graph is Breadth first traversal

→ Complexity = $O(V+E)$

All weights are = 1

Floyd Warshal Algo

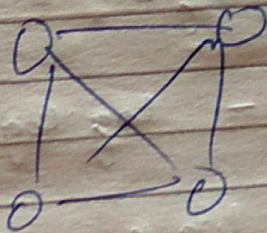
{ All pair shortest path algo }
 Saare pair k beech nikalte

↓
 Dynamic Prog problem

Comp = $O(V^3)$

(CS13)
17Complexity of Bellman Ford algo on complete graph of n vertices

Complete graph -

has n vertices, every
vertex is
connected to

$$\text{No of edges} = \frac{n(n-1)}{2} = E$$

$$\begin{aligned} \text{Complexity} &= (VE) \\ &= \left[\frac{n(n-1)}{2} \right] n \\ &= O(n^3) \end{aligned}$$

(CS12)
21worst case time to search element
in balanced binary search tree with
 $n \times 2^n$ elements

$$\text{AVL ki comp} = \log n$$

$$\begin{aligned} &= \log(n 2^n) = \log n + \log 2^n = \\ &= n + \log n \\ &= O(n) \end{aligned}$$

(CS13)
41no of elements that can be sorted in $O(\log n)$ time
using heap sort

$$\text{Complexity of heap sort was } n \log n$$

$$n' \log n' = \log n$$

$$d = (\log n)$$

$$a \quad O(1)$$

$$b \quad O(\sqrt{\log n})$$

$$c \quad O\left(\frac{\log n}{\log(\log n)}\right)$$

$$\frac{\log(n \log n)}{\log \log(n \log n)}$$

$$= \log \left[\frac{\log n}{\log \log n} \right]$$

$$\frac{\log n}{\log \log n} \cdot \left[\log \log n - \log \log \log n \right]$$

$$= \log n - \frac{\log n \log \log \log n}{\log \log n}$$

CS13
19

tightest upper bound that represent no of swap
 required to sort n numbers using selection sort
 Per pass 1 swap lgt is n
 $\Rightarrow (n-1)$ swap = $O(n)$

CS12
8

Let $w(n)$ & $A(n)$ are worst & avg case complexity
 which is always true

A) $A(n) = \Theta(w(n))$

B) $A(n) = \Theta(w(n))$

☒ C) $A(n) = O(w(n))$

D) $A(n) = o(w(n))$

CS15
Set 3
35

$f(n) = n$ $g(n) = n^{1+\sin n}$
 Relⁿ b/w $f(n)$ & $g(n)$

kahi $f(n)$ bda kahi $g(n)$ bda
 $\Rightarrow$ Neither Θ nor $\rightarrow$

optimal substructure - If an optimum solⁿ of problem is possible when we find optimum solⁿ of subproblem

eg shortest path algo exhibit property of optimal substructure

Min dist from A to X via B & C can be found only when we find min (AB, BC, CX)

But this won't work at Longest path algo (Longest path without cycling)

longest path (A-D) = A-B-C-D
but longest path (A-B) = A-E-D-C-B

Therefore longest path algo do not exhibit optimum substructure

CS14
Set2
41

CS10
12

$$A = 0.0001 n^2 \text{ time units}$$

$$B = 10 n \log_{10} n \text{ time units}$$

n for which B is preferred over A

$$\text{So } \text{Time}(B) < A$$

$$A = 12$$

$$b = 10$$

$$c = 6$$

$$d = 5$$

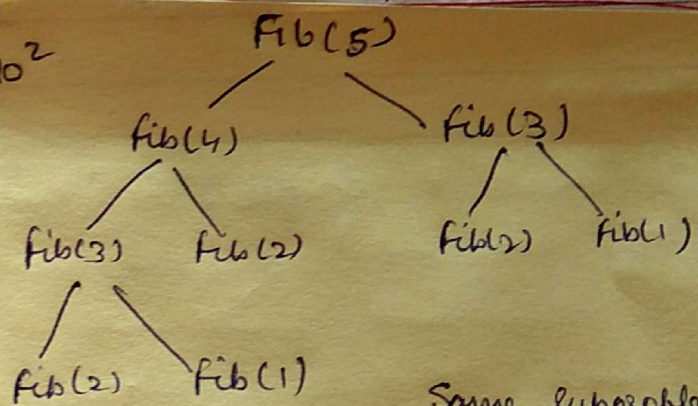
$$A = 10^{-4} \times 12^2$$

$$B = 10^{-4} \times 10^2$$

$$C =$$

$$D = 10^{-4} \times$$

MERGE SORT & QUICK SORT ARE NOT DYNAMIC PROGRAM - They are Divide & Conquer
↳ subproblems are not overlapping



Same subproblem is used for sol of other sub-subproblems

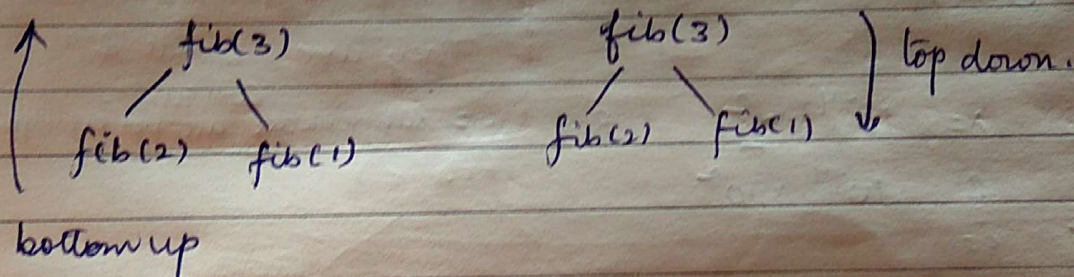
Overlapping subproblems.

to final solⁿ, but this we select subproblems which give optimum results.

* Dynamic programming overlapping b/w no solⁿ h, element can be common b/w some subproblems
* whereas in divide & conquer, no element can be common in 2 subproblems.

Memoization $\Rightarrow$ Jisse bhi subproblem dynamic programming solve kr rai h uske results ek data str me save krte hote h, whenever result of subprogram is required, it is fetched from there.

Dynamic programming uses trees. Can solve in top-down fashion / bottom up fashion.



In divide & conquer, we know where to divide into subproblem (eg merge sort) but in dynamic, we don't know where problem will break into its subproblem. The subproblem which produce optimum result will be broken down further.

- * Matrix chain multiplication
 - * LCS (longest common subsequence)
 - * Subset sum problem
- } Problems of DP

Matrix chain Multiplication

Multiply $A_{3 \times 2}$ $B_{2 \times 3}$ $C_{3 \times 1}$ $D_{1 \times 2}$ this with min no of operations.

if $A_{3 \times 2}$ $B_{2 \times 3}$ C

$A_{m \times n}$ $B_{n \times k}$ is to be multiplied then for normal procedure $m \times n \times k$ multiplications are required

For a subproblem of size 4 start solving by

$((AB)C)D$

$(A(BC))D$

$((AB)(CD))$

$A(B(CD))$

$A(B(CD))$

only 1 multiplication

cost of subproblem of size 1 = 0

2

A B C D
3x2 2x3 3x1 1x2

$$\left[\begin{array}{l} AB = 3 \times 2 \times 3 = 18 \quad CD = 3 \times 1 \times 2 = 6 \\ BC = 2 \times 3 \times 1 = 6 \end{array} \right]$$

Subproblem of size 3 = $\min \{ AB \times C = 18 + 3 \times 3 \times 1 = 27$

$\left\{ \begin{array}{l} A \times BC = 3 \times 2 \times 1 + 6 = 12 \end{array} \right.$

$\left\{ \begin{array}{l} B \times CD = 2 \times 3 \times 2 + 6 = 18 \end{array} \right.$

$\left\{ \begin{array}{l} BC \times D = 6 + 2 \times 1 \times 2 = 10 \end{array} \right.$

cost of subproblem of size 4 = $\min \{ \begin{array}{l} ABC \times D = 12 + 3 \times 1 \times 2 = 18 \\ A(BCD) = 3 \times 2 \times 2 + 10 = 22 \\ (AB) \times (CD) = 18 + 6 + 3 \times 3 \times 2 = 42 \end{array} \right.$

ABC can be multiplied in 2 ways

[ABC & D are matrices]

(AB)C or A(BC)

for ABCD $\Rightarrow ((AB)C)D$ or $(A(BC))D$ or $(AB)(CD)$
or $A(BC)D$ or $A(B(CD))$

Eg $(A_1)_{p_0 \times p_1} \quad (A_2)_{p_1 \times p_2} \quad (A_3)_{p_2 \times p_3} \quad (A_4)_{p_3 \times p_4} \dots$

we have to multiply this chain

Create a table we have 4 matrices so

table should be 4x4

ABC * D

A * BC * D

Most appropriate

Rajshree
PAGE NO.:
DATE:

	$\hat{j} = 1$	$\hat{j} = 2$	$\hat{j} = 3$	$\hat{j} = 4$
$\hat{i} = 1$	0			
$\hat{i} = 2$	x	0		
$\hat{i} = 3$	x	x	0	
$\hat{i} = 4$	x	x	x	0

Vikram Samvat 2071
Phalgun Shukla 6
Shak Samvat 1936
Phalgun 5

is lie diagonal entries sub O has
For filling rest of cells. use:

$$m[i, j] = \min_{i \leq k < j} \{ m[i, k] + m[k+1, j] \} + p_i \cdot p_{k+1} \cdot p_j$$

$$\min [i, j] = 0 \quad \& \quad i = j$$

Q A 3×2 B 2×1 C 1×2 D 2×2
 $P_0 P_1$ $P_1 P_2$ $P_2 P_3$ $P_3 P_4$

Find min no of multi

$$m[1,2] = \min_{1 \leq k < 2} \left\{ \underset{\substack{\downarrow \\ 0}}{m[1,1]} + \underset{\substack{\downarrow \\ 0}}{m[2,2]} + \underset{\substack{\downarrow \\ 3 \quad 2 \quad 1}}{p_0 \times p_1 \times p_2} \right\}$$

x can be

1 ko 1 se
mul hi
na krna

2 to 2
se mul
hi ni tme

$$\Rightarrow m[1, 2] = 6$$

we could have directly calculated $m[1,2]$

$$m[1, 2] = A_{3 \times 2} \times^0 B_{2 \times 1}$$

$$3 \times 2 \times 1 = 6$$

$$A_1 \times A_2 \times A_3 \quad \dots \quad A_n$$

$\swarrow \quad \downarrow \quad \quad \quad \downarrow$
 $p_0, p_1 \quad p_2, p_3 \quad \quad \quad p_{n-1}, p_n$

breaking it into subproblem

$$A_i \times A_{i+1} \quad \dots \quad A_j$$

Solving it

$$A_i \times (A_{i+1} \quad \dots \quad A_j) \quad \text{way 1}$$

$$\text{cost} = m[i, i] + m[i+1, j] + p_{i-1} \times p_i \times p_j$$

$$\text{way } (A_i \times A_{i+1}) \times (A_{i+2} \quad \dots \quad A_j) \quad \text{way 2}$$

$$\text{cost} = m[i, i+1] + m[i+2, j] + p_{i-1} \times p_i \times p_j$$

we need to find min of all these ways.

$$m[i, j] = \begin{cases} 0 & \text{if } i=j \\ m[i, k] + m[k+1, j] + p_{i-1} \times p_k \times p_j & \text{if } i < j \end{cases}$$

कामना को संतुष्ट नहीं करना अच्छा है।

It is well not to satisfy one's desire.

2015

$i < k < j$

25

FEBRUARY
WEDNESDAY
Week 09 (056-309)

फरवरी
बुधवार
सप्ताह ०९ (०५६-३०९)

सूर्योदय ६:२० Sun Rise 6:20

Raishree

Vikram Sankranti २०७९
Phalgun Sankranti १९
Shak Sankranti १६३६
Phalgun Sankranti ६

सूर्यास्त ५:४० Sun

$A \times B \times C \times D$
 $3 \times 1 \quad 1 \times 2 \quad 2 \times 2 \quad 2 \times 4$

	1	2	3	4
4			16	0
3	10	4	0	
2	6	0		
1	0			

$k=1$
 $m[1,1] + m[2,3] + 3 \times 1 = 10$

$m[1,3] = \min$

Matrix 1 x matrix 2

$k=2$
 $m[1,2] + m[3,3] + 3 \times 2 \times 2 = 6 + 0 + 12 = 18$

$\Rightarrow m[1,3] = 10$ Ans

when $A \times B \times C$ is multiple

2015

श्रद्धा से मनुष्य क्या नहीं कर सकता? सब कुछ कर सकता है।
What can a person not accomplish by faith? He can do everything.

$A_1 \times A_2 \times A_3 \times A_4 \dots A_n$

$\downarrow \quad \downarrow \quad \downarrow$
 $P_0, P_1 \quad P_1, P_2 \quad P_2, P_3$

Page No.
 DATE
 P_{n-1}, P_n

⇒ table will be

0	6	12	
x	0	4	
x	x	0	4
x	x	x	0

Now let's find

$$m[1,3] = \min_{1 \leq k \leq 3} \{ m[1,1] + m[2,3] + P_0 P_1 P_3 \}$$

k can be 1 or 2 or 3

$$= \min \begin{cases} \text{when } k=1 & m[1,1] + m[2,3] + P_0 P_1 P_3 = 16 \\ \text{when } k=2 & m[1,2] + m[3,3] + P_0 P_2 P_3 = 12 \end{cases}$$

Hence $m[1,3] = 12$

$$m[2,4] = \min_{2 \leq k \leq 4} \{ \dots \}$$

k can be 2 or 3

$$\begin{cases} k=2 & m[2,2] + m[3,4] + P_1 P_2 P_4 = 8 \\ k=3 & m[2,3] + m[4,4] + P_1 P_3 P_4 = 12 \end{cases}$$

⇒ $m[2,4] = 8$

$$m[1,4] = \min_{1 \leq k \leq 4} \{ \dots \}$$

k can be 1 or 2 or 3

$$\min \begin{cases} k=1; & m[1,1] + m[2,4] + P_0 P_1 P_4 = 20 \\ k=2; & m[1,2] + m[3,4] + P_0 P_2 P_4 = 16 \\ k=3; & m[1,3] + m[4,4] + P_0 P_3 P_4 = 24 \end{cases}$$

$$\Rightarrow m[1, 4] = 16$$

Ans

which came from

$$m[1, 2] + m[3, 4] + P_0 P_2 P_4$$

$$\uparrow$$

$$(A \times B) \times (C \times D)$$

will compute lowest Ans

II problem of dynamic programming

Longest Common Subsequence

$X = A B B C B A C B$

$Y = A C B B C A C A B$

Substring me hum beech k char skip nai kr
Subsequence me kr sakte h.

eg $A B B C C D$

$A B C D$ is a subsequence

$A B B C B A C B$

$A C B B C A C A B$

$\Rightarrow$ subsequence length
 $= 7$

Q $X = A B C B A C B$
 $x = A B B C B A C \leftarrow \text{length} = m$
 $Y = A C B C B A C \leftarrow \text{length} = n$

Then create a table of $(m+1) \times (n+1)$ size

$$\text{Length}(X_m, Y_n) = \begin{cases} 0 & \text{if } m=0 \text{ or } n=0 \\ \max[L(X_{n-1}, Y_m), L(X_n, Y_{m-1})] & \text{if } x_n \neq y_m \\ L(X_{n-1}, Y_{m-1}) + 1 & \text{if } x_n = y_m \end{cases}$$

Vikram Samvat 2071
Chaitra Krishna 8
Shak Samvat 1936
Phalgun 23

विक्रम सम्वत् २०७१
चैत्र कृष्ण ८
शक सम्वत् १९३६
फाल्गुन २३

Choice मार्च 14
शनिवार SATURDAY
सप्ताह ३१ (०७ मार्च २०२३)
Subsequence possible

सूर्योदय ६:०६ Sun Rise 6:06

सूर्यास्त ५:५४ Sun Set 5:54

X = A B C B A A C Y = B A C B A A B C

	O	A	B	C	B	A	A	C
O	0	0	0	0	0	0	0	0
B	0	← 0	← 1	← 1	← 1	← 1	← 1	← 1
A	0	← 1	← 1	← 1	← 1	← 2	← 2	← 2
C	0	↑ 1	← 1	← 2	← 2	← 2	← 2	← 3
B	0	↑ 1	← 2	← 2	← 3	← 3	← 3	← 3
A	0	↑ 1	↑ 2	← 2	↑ 3	← 4	← 4	← 4
A	0	↑ 1	↑ 2	← 2	↑ 3	← 4	← 5	← 5
B	0	↑ 1	↑ 2	← 2	↑ 3	← 4	↑ 5	↑ 5
C	0	↑ 1	↑ 2	↑ 3	← 3	↑ 4	↑ 5	↑ 6

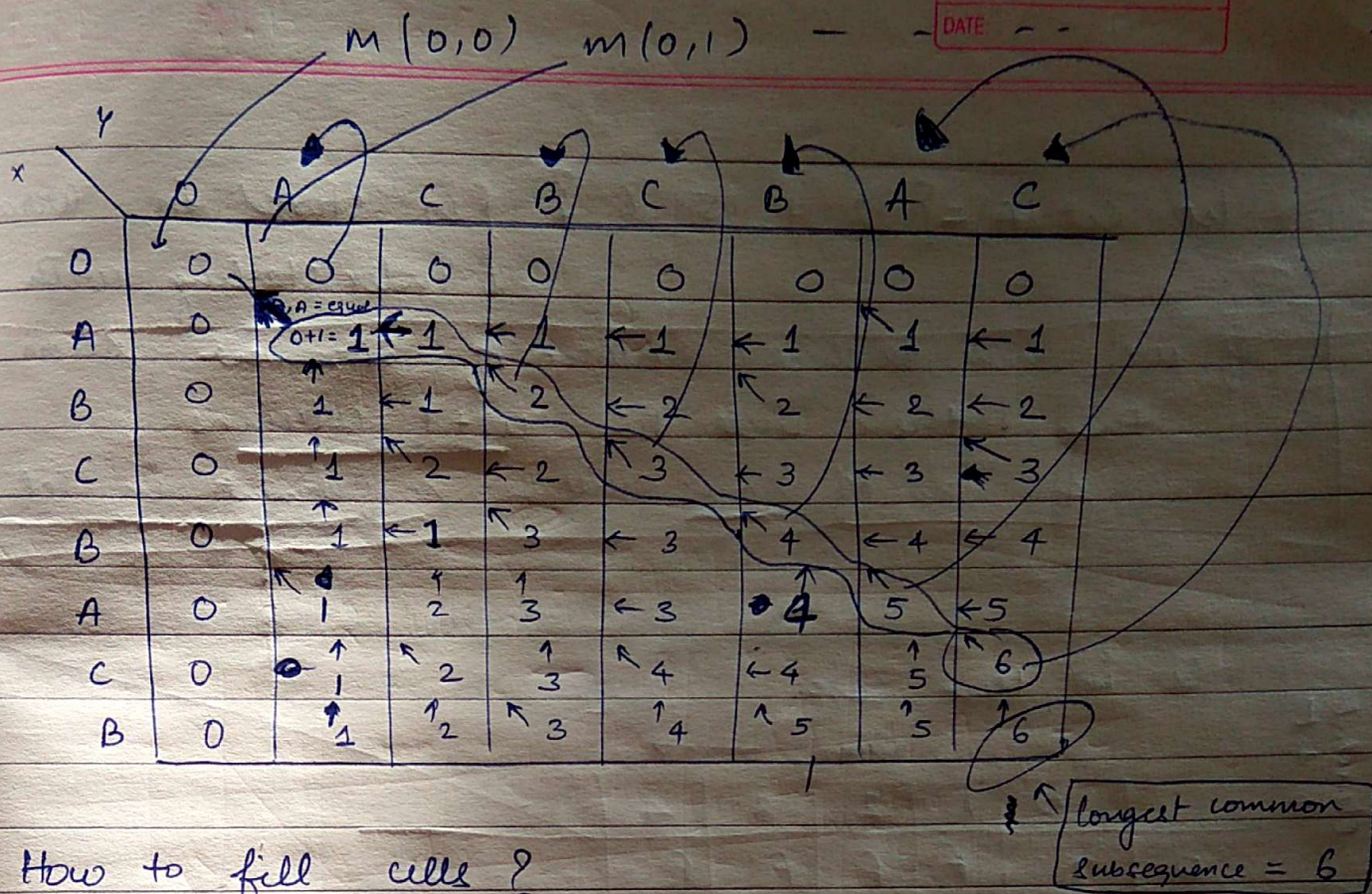
ACBAAC

* If row head $\neq$ column head then cell = max [above, left]
if above elem $\neq$ left elem put both arrows
If row head = column head then cell = [1 + diagonal up val]

We can compute table rowwise or columnwise
निम्न को छूट जाना कैसा खतरनाक है।
How dangerous it is to fall out of routine.
Columnwise

Subset Sum problem

we are having set of +ve integers. We have to find a subset whose sum is...



How to fill cells?

1* $M[i, j] = 0$ when either $i=0$ or $j=0$

2* $M[i, j] = \max [M[i-1, j], M[i, j-1]]$ if $x_i \neq y_j$

$$m[1, 2] = \max (m[0, 2], m[1, 1])$$

3* $M[i, j] = \max (M[i-1, j-1] + 1)$ if $x_i = y_j$

The elements which have  arrow are part of longest common subsequence

The elements with $\leftarrow, \uparrow$ are to be excluded in longest subsequence
 $\Rightarrow$ seq is A B C B A C

III prob of dynamic prog

Subset Sum problem

Set = unordered collection of DISTINCT elements

we are having set of +ve integers. We have to find if there is a subset present whose sum is W ?

If our set contains 0 or -ve integers then this prob is NP hard $\Rightarrow$ can't be solved.

Let Set $(A) = \{ 2 \ 1 \ 5 \ 6 \}$ { is stored in array
index starting with 1 }

If we have to check subset of size ~~sum~~ sum w
then there will be $w+1$ columns
4 no of rows will be no of elements in set + 1

So in this case

first i th element of set

		Sum →				
		0	1	2	3	4
0	T	F	F	F	F	
1	T	F	T			
2	T		T			
3	T		T			

cell (i, j) is true if first i elements of set can have a sum $= j$

taking element (subset) = ϕ

for ($i=1$; $i \leq n$; $i++$)

{

for ($j=1$; $j \leq w$; $j++$)

{

$S[i, j] = S[i-1, j] || (A[i] \leq j \ \&\& \ S[i-1][A[i]-1])$

Koi cell True h to uske niche wali entry bhi true hi hogi

$S[1, 1] = S[0, 1] || [A[1] \leq 1 \ \&\& \ \text{---}] = \text{false}$
false

$$S[1,2] = S[0,2] \because [A[1] \leq 2 \text{ \& } S[0,2-2]] = T$$

If $S(n, w) = T$ then there exist a subset whose value is w

Floyd warshall Algo / All pair shortest path algo
ki complexity poochi jati hai $\Rightarrow O(n^3)$
where n = no of vertices

Subset sum problem

find if sum = 6 is possible in subset of

Set $\{2, 3, 4\}$

↑ ↑ ↑
elem1 elem2 elem3

$a[] = \{2, 3, 4\}$

Shows sum

		0	1	2	3	4	5	6
0	T	F	F	F	F	F	F	F
1	T	F	T	F	F	F	F	F
2	T	F	T	T	F	T	F	F
3	T	F	T	T	T	T	T	T

means first 2 elements

$$S[i, j] = \begin{cases} T & \text{if } j=0 \\ F & \text{if } j \neq 0 \text{ \& } i=0 \\ S[i-1, j] \text{ US } S[i-1, j-a[i]] & \text{if } i \neq 0 \text{ \& } j \neq 0 \end{cases}$$

Coz agar first 2 elements se sum mil gya to first 3 elements ki sum se bhi milaga hai
phir row true to current row true

Previous row se koi element mila hai jo current element to subset me hi mila hai